MLPerf Storage v3.0 — System Description

potato8-qlc-np2-closed-tcm · CLOSED division · submitter: farmgpu

System name	potato8-qlc-np2-closed-tcm
Submitter	FarmGPU (farmgpu.com)
Division	CLOSED
Benchmark	KVCache — MLPerf Storage v3.0 (mlpstorage 3.0.37)
Solution type	local-storage (direct-attached / integrated client storage)
Availability	available
Usable capacity	21461.7986 TiB
Deployment	on-premises, bare metal
Rack units	0 U storage-under-test (integrated in client nodes; clients excluded per the MLPerf RU definition). Physical deployment footprint: 8 x 2U = 16 U of client nodes.
Client topology	8 node(s) x 2 MPI rank(s)/node = 16 client rank(s)
Model(s) exercised	llama3.1-8b (Opt 1, 2); llama3.1-70b-instruct (Opt 3)
Run date(s)	2026-07-08 / 2026-07-09
Document prepared	2026-08-03

1. System overview

The storage under test is client-local NVMe: 8 identical bare-metal node(s), each benchmarking its own independent 24-drive Solidigm™ D5-P5336 QLC array (md RAID-0, 512 KiB chunk, XFS at /mnt/qlc). There is no external or shared storage system and no data sharing between nodes; the 400GbE fabric carries only MPI coordination and result-file aggregation, not benchmark I/O. Aggregate results scale with client count (Rules §6.3.3). Measured usable capacity is 21461.7986 TiB across 8 node(s), taken from the mounted filesystem rather than computed from drive counts.

2. Hardware

Chassis	Supermicro SYS-212H-TN (2U)
CPU	1x Intel® Xeon® 6781P (80 cores / 160 threads)
DRAM	512 GiB installed — 8 x 64 GB DDR5-5600 (Micron MTC36F2046S1PC56BG1); 503 GiB available to the OS
Storage under test	24x Solidigm™ D5-P5336 122.88 TB QLC PCIe Gen4 NVMe (P/N SBFPF2BV0P12); single md RAID-0 (512 KiB chunk) + XFS, mounted at /mnt/qlc
Data NIC	2 × NVIDIA® ConnectX®-7 400GbE (RoCE-capable)
Management NIC	1 × 1 GbE
Power supplies	2x 1600 W Titanium (208 V); min PSUs active = 1 (1 redundant)

3. Software

Operating system	Ubuntu 24.04
Benchmark	MLCommons mlpstorage 3.0.37, kv_cache workload
Storage stack	Linux md RAID-0 (512 KiB chunk) + XFS
Launcher	Open MPI (mpirun)
Memory allocator	tcmalloc (libtcmalloc_minimal.so.4)

4. Settings and non-default configuration

• tcmalloc (libtcmalloc_minimal.so.4) as the client-side memory allocator, injected via LD_PRELOAD through mpirun (--mpi-params). This is the only client tuning applied.

• --skip-validation was used on this multi-host run. The results directory is genuinely shared across all 8 hosts; .fs_probe/*.ok sentinels are present in the submitted run directories as evidence.

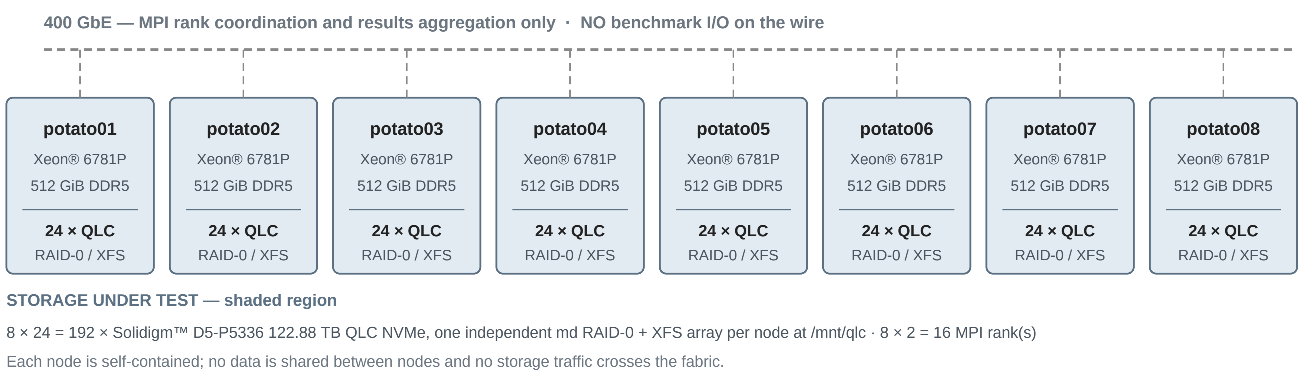
• 2 MPI ranks per host (16 ranks total). Reported capacity and throughput aggregate 8 independent single-node arrays.

5. Power requirements

System component	Power supply unit	Nameplate rated power	Design power
Client/storage node x8	PSU 1	1600 W	1600 W
	PSU 2 (redundant)	1600 W	—
Totals		8 x 3200 W	8 x 1600 W

Design power is stated at one active PSU per node (redundant configuration, min PSUs active = 1); nameplate totals count both PSUs. Client-network switching is excluded per §11.4.2. Measured wall power was not captured; values are nameplate-derived and represent provisioned rather than consumed power.

6. System topology



Each of the 8 node(s) is self-contained: CPU + 512 GiB installed DDR5 + a local 24x QLC md-RAID0/XFS array (the storage under test). No storage traffic crosses the fabric; the 400GbE links carry MPI coordination and NFS result aggregation only.

7. Rack units

Per the MLPerf Storage RU definition, total_rack_units counts the storage-under-test plus any required backend gear and excludes client nodes. This solution integrates its storage into the client nodes with no dedicated storage chassis or backend switching, so total_rack_units = 0 (the systems YAML records 0, which the submission validator confirms). For physical context, the deployment occupies 8 x 2U = 16 U of client-node rack space.

Generated for the MLPerf Storage v3.0 farmgpu submission from systems/potato8-qlc-np2-closed-tcm.yaml. Hardware values correspond to the system YAML; capacity is as measured. Prepared 2026-08-03.